

Lecture 11: 4NF and Multivalued Dependencies

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Beyond BCNF

The Problem with BCNF

Even if a relation is in BCNF, it can still have data redundancy.

Example: A relation that stores courses, the teachers for that course, and the required textbooks. Assume teachers and textbooks are independent (any teacher for a course uses all its textbooks).

CourseInfo		
course	teacher	textbook
CSL303	Prof. Smith	Databases 101
CSL303	Prof. Smith	SQL Fundamentals
CSL303	Prof. Jones	Databases 101
CSL303	Prof. Jones	SQL Fundamentals

The Problem with BCNF

CourseInfo		
course	teacher	textbook
CSL303	Prof. Smith	Databases 101
CSL303	Prof. Smith	SQL Fundamentals
CSL303	Prof. Jones	Databases 101
CSL303	Prof. Jones	SQL Fundamentals

- **Redundancy:** The fact that 'Databases 101' is a textbook for CSL303 is stored twice (once for each teacher).
- This relation is in BCNF because there are no non-trivial FDs. The only key is {course, teacher, textbook}.
- The problem is a different kind of dependency: a **multivalued dependency**.

Multivalued Dependencies (MVDs)

Definition of Multivalued Dependency

A multivalued dependency (MVD) is a constraint that requires that certain tuples be present in a relation.

Definition: A multivalued dependency (MVD) is denoted as $\alpha \twoheadrightarrow \beta$. It means that the values of attributes in α determine a *set* of values for attributes in β , and this set is independent of the other attributes in the relation.

Definition of Multivalued Dependency

In our example:

- For the course 'CSL303', the set of teachers is {'Prof. Smith', 'Prof. Jones'}.
- For the course 'CSL303', the set of textbooks is {'Databases 101', 'SQL Fundamentals'}.

The MVDs are:

- $\text{course} \twoheadrightarrow \text{teacher}$
- $\text{course} \twoheadrightarrow \text{textbook}$

This means the set of teachers for a course is independent of the set of textbooks for that course.

Fourth Normal Form (4NF)

4NF is designed to eliminate the redundancy caused by MVDs.

Definition: A relation is in 4NF if for every non-trivial MVD $\alpha \twoheadrightarrow \beta$, α is a superkey for the relation.

- A non-trivial MVD $\alpha \twoheadrightarrow \beta$ is one where:
 1. β is not a subset of α .
 2. $\alpha \cup \beta$ is not all the attributes in the relation.

Our CourseInfo relation is **not** in 4NF because course is not a superkey.

Solution: Decompose the relation.

- R1(course, teacher)
- R2(course, textbook)

Both new relations are in 4NF.

3NF, BCNF, and 4NF Examples

Example 1: 3NF, not BCNF

- Relation: Enroll(student, course, professor)
- FDs: {student, course} $\rightarrow$ professor, professor $\rightarrow$ course
- Not in BCNF because professor is not a superkey.

3NF, BCNF, and 4NF Examples

Example 1: 3NF, not BCNF

- Relation: Enroll(student, course, professor)
- FDs: {student, course} $\rightarrow$ professor, professor $\rightarrow$ course
- Not in BCNF because professor is not a superkey.

Example 2: BCNF, not 4NF

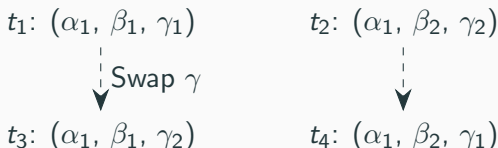
- Relation: CourseInfo(course, teacher, textbook)
- FDs: None. MVDs: course $\twoheadrightarrow$ teacher — textbook
- In BCNF (no FD violations), but not in 4NF because course is not a superkey.

Discovering MVDs

Algorithm for Discovering MVDs

To check if an MVD $\alpha \twoheadrightarrow \beta$ holds in a relation instance, we can use the following algorithm:

1. Find all pairs of tuples, t_1 and t_2 , that agree on all attributes in α .
2. For each such pair, let the remaining attributes be $\gamma = R - \alpha - \beta$.
3. The MVD holds only if two other tuples, t_3 and t_4 , also exist in the relation for every pair:
 - t_3 must match t_1 on α and β , and t_2 on γ .
 - t_4 must match t_2 on α and β , and t_1 on γ .



Discovery Algorithm: Example Run 1

Check: $\text{course} \twoheadrightarrow \text{teacher}$

Let $\alpha = \{\text{course}\}$, $\beta = \{\text{teacher}\}$, $\gamma = \{\text{textbook}\}$.

course	teacher	textbook
t_1 : CSL303	Prof. Smith	Databases 101
t_2 : CSL303	Prof. Jones	SQL Fundamentals
t_3 : CSL303	Prof. Smith	SQL Fundamentals
t_4 : CSL303	Prof. Jones	Databases 101

- t_1 and t_2 agree on 'course'.
- We need to check if t_3 and t_4 exist.
- t_3 exists in the table (row 2 of the original example).
- t_4 exists in the table (row 3 of the original example).
- Since this holds for all pairs, the MVD $\text{course} \twoheadrightarrow \text{teacher}$ holds.

Discovery Algorithm: Example Run 2

Check: $\text{teacher} \twoheadrightarrow \text{course}$

Let $\alpha = \{\text{teacher}\}$, $\beta = \{\text{course}\}$, $\gamma = \{\text{textbook}\}$.

Assume we have a different relation:

teacher	course	textbook
t_1 : Prof. Smith	CSL303	Databases 101
t_2 : Prof. Smith	CSL101	Intro to C++

- t_1 and t_2 agree on 'teacher'.
- We must check if a tuple $t_3 = (\text{'Prof. Smith'}, \text{'CSL303'}, \text{'Intro to C++'})$ exists.
- If this tuple is not in the relation, the MVD **does not hold**.

The Chase Algorithm for FDs

The Chase is a fundamental algorithm used to reason about dependencies.

Primary Usage: To determine if a set of FDs F logically implies another FD $\alpha \rightarrow \beta$.

Algorithm Idea:

1. Create a "tableau" (table) with two rows, representing two tuples that agree on α .
2. Use non-distinguished variables for attributes where they differ.
3. Repeatedly apply the FDs in F . If an FD $\gamma \rightarrow \delta$ applies to the tableau, and two rows agree on γ , force them to agree on δ by equating their values.
4. If, after applying all FDs, the two rows are forced to agree on β , then the dependency holds.

Chase Example: Does $\{A \rightarrow B, B \rightarrow C\}$ imply $A \rightarrow C$?

1. **Initial Tableau:** Two rows agree on A, but differ on C.

A	B	C
a_1	b_1	c_1
a_1	b_2	c_2

2. **Apply $A \rightarrow B$:** The two rows agree on A (a_1), so they must agree on B. We set $b_2 = b_1$.

A	B	C
a_1	b_1	c_1
a_1	b_1	c_2

Chase Example: Does $\{A \rightarrow B, B \rightarrow C\}$ imply $A \rightarrow C$?

2. Apply $A \rightarrow B$: The two rows agree on A (a_1), so they must agree on B. We set $b_2 = b_1$.

A	B	C
a_1	b_1	c_1
a_1	b_1	c_2

3. Apply $B \rightarrow C$: Now the two rows agree on B (b_1), so they must agree on C. We set $c_2 = c_1$.

A	B	C
a_1	b_1	c_1
a_1	b_1	c_1

Conclusion: The chase forced the two rows to agree on C. Therefore, the dependency $A \rightarrow C$ is logically implied.

Extending the Chase for MVDs

The Chase algorithm can be extended to infer dependencies involving MVDs.

Recap FD Rule: If we have an FD $\alpha \rightarrow \beta$, and two rows agree on α , we force them to agree on β .

New MVD Rule: If we have an MVD $\alpha \twoheadrightarrow \beta$, and two rows t_1, t_2 agree on α , we generate a **new row** t_3 .

- t_3 takes its α and β values from t_1 .
- t_3 takes its other attribute values from t_2 .

Extending the Chase for MVDs

Example: Test if $\{A \twoheadrightarrow B, B \rightarrow C\}$ implies $A \rightarrow C$.

Initial Tableau:

A	B	C
a_1	b_1	c_1
a_1	b_2	c_2

Apply $A \twoheadrightarrow B$: t_1, t_2 agree on A. Generate new row t_3 . t_3 takes

A,B from t_1 and C from t_2 .

A	B	C
a_1	b_1	c_1
a_1	b_2	c_2
a_1	b_1	c_2

Apply $B \rightarrow C$: t_1 and t_3 agree on B (b_1). Force C to be same.

A	B	C
a_1	b_1	c_1
a_1	b_2	c_2
a_1	b_1	c_1

Thank You!
Questions?